EXPERIMENT 23

Text Coherence Evaluation using Lexical Overlap and Transition

AIM

To develop a Python program that evaluates the semantic and lexical coherence of a given text by measuring adjacent sentence continuity.

PROCEDURE

1. Split the text into sequential sentences.
2. Preprocess and remove stopwords to extract content words.
3. Compute sentence-to-sentence lexical overlap: $Score(S_i, S_{i+1}) = \frac{|Tokens(S_i) \cap Tokens(S_{i+1})|}{|Tokens(S_i) \cup Tokens(S_{i+1})|}$.
4. Compute average discourse continuity score across all sentence transitions.
5. Compare coherent text against an incoherent/scrambled passage.

PROGRAM

```
import re

stopwords = {'the', 'is', 'at', 'which', 'on', 'a', 'an', 'in', 'and', 'or', 'to',
'of', 'for', 'it'}

def evaluate_coherence(text):
    sentences = [s.strip() for s in text.split('.') if len(s.strip()) > 5]
    if len(sentences) < 2:
        return 1.0, []

    sentence_words = []
    for s in sentences:
        words = set(re.findall(r'\b[a-z]{3,}\b', s.lower())) - stopwords
        sentence_words.append(words)

    overlaps = []
    for i in range(len(sentence_words) - 1):
        s1, s2 = sentence_words[i], sentence_words[i+1]
        intersection = len(s1 & s2)
        union = len(s1 | s2)
        jaccard = intersection / union if union > 0 else 0.0
        overlaps.append((i+1, i+2, jaccard, s1 & s2))

    avg_coherence = sum(x[2] for x in overlaps) / len(overlaps)
    return avg_coherence, overlaps

coherent_text = """Machine learning relies heavily on large datasets. These datasets
provide the training samples required for statistical models. Statistical models
learn latent representations to perform automated predictions."""
```

```
incoherent_text = """Astronauts launched into space yesterday. Pizza remains one of
the most popular Italian foods worldwide. Quantum mechanics investigates the
behavior of subatomic particles."""

print("=== Evaluating Coherent Text ===")
score1, details1 = evaluate_coherence(coherent_text)
print(f"Overall Coherence Score: {score1:.4f}")
for s1, s2, sc, shared in details1:
    print(f"  Sentence {s1} -> {s2}: Overlap={sc:.4f} | Shared Vocab: {shared}")

print("\n=== Evaluating Incoherent Text ===")
score2, details2 = evaluate_coherence(incoherent_text)
print(f"Overall Coherence Score: {score2:.4f}")
for s1, s2, sc, shared in details2:
    print(f"  Sentence {s1} -> {s2}: Overlap={sc:.4f} | Shared Vocab: {shared}")
```

OUTPUT

```
> python3 experiment_23.py
---------------------------------------------------------------------------------------------------
=== Evaluating Coherent Text ===
Overall Coherence Score: 0.2361
  Sentence 1 -> 2: Overlap=0.2500 | Shared Vocab: {'datasets'}
  Sentence 2 -> 3: Overlap=0.2222 | Shared Vocab: {'models', 'statistical'}

=== Evaluating Incoherent Text ===
Overall Coherence Score: 0.0000
  Sentence 1 -> 2: Overlap=0.0000 | Shared Vocab: set()
  Sentence 2 -> 3: Overlap=0.0000 | Shared Vocab: set()
[Process completed with exit code 0]
```

RESULT

Thus, the program evaluating text coherence through inter-sentence lexical overlap and discourse continuity was successfully implemented.